

SHILATE

Software Hardware In Loop
Automotive Testing Environment

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The Industry Shift

- Vehicles are becoming **software-defined** — behaviour updated over-the-air
- Reinforcement learning (RL) is maturing as a practical control tool
- But no open-source pipeline bridges **RL training** and **SDV deployment**

Four Open Challenges

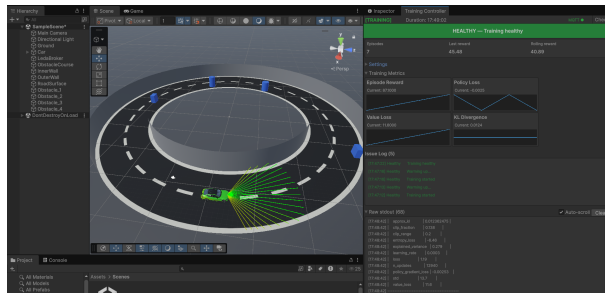
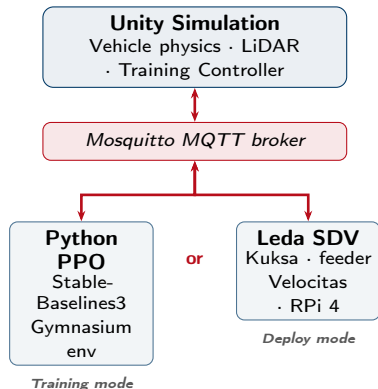
- 1 Simulation $\leftrightarrow$ middleware protocol coupling
- 2 Standardised signal representation (VSS)
- 3 Operational safety monitoring of learned policies
- 4 Reproducible, one-click training workflow

Goal

Build an **end-to-end, open-source** pipeline that:

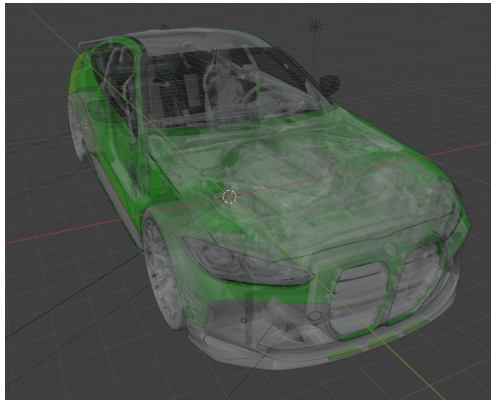
- Trains a deep RL agent in a Unity physics sim
- Deploys the policy onto a real Eclipse Leda SDV stack
- Uses the **same MQTT / VSS layer** from training to hardware

System Architecture — Training vs. Deploy Mode



Simulation environment overview

- **Training:** Unity Sim ↔ MQTT ↔ Python PPO
- **Deploy:** Unity Sim ↔ MQTT ↔ Eclipse Leda SDV
- Same **env0/** topic namespace in both modes — no protocol changes at deploy time



Rear-wheel-drive vehicle model

Vehicle & Sensor

- 1 500 kg RWD chassis — four WheelColliders, Pacejka tire model
- 21-ray LiDAR spanning 120° at 50 m range, published at 10 Hz
- Observation $o \in \mathbb{R}^{23}$: rays + normalised speed + steer

Reward Function

$$r_t = \underbrace{2 \Delta p_t}_{\text{progress}} + \underbrace{(-5) 1_{\text{collision}}}_{\text{collision}} + \underbrace{+50 1_{\text{lap}} + +25 1_{\frac{1}{2}\text{-turn}}}_{\text{milestones}}$$

Training Controller (Editor window)

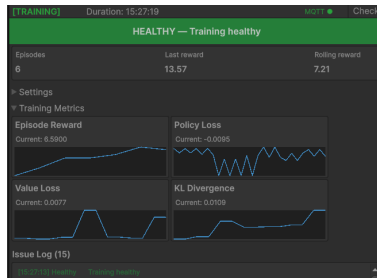
- One-click **Start Training / Run Model / Stop**
- Live health banner — 5 failure modes monitored
- Real-time reward, loss, and KL-divergence graphs

Algorithm & Architecture

- Proximal Policy Optimisation (PPO)
- Actor-Critic MLP: two hidden layers [256, 128]
- Action space: $a = [\delta, \theta, \beta]$ — $\text{steer} \in [-1, 1]$, $\text{throttle} \in [0, 1]$, $\text{brake} \in [0, 1]$

Key Hyperparameters

Parameter	Value
Total timesteps	100 000
Learning rate	3×10^{-4}
Rollout steps	2 048
Batch size	64
Clip ϵ / γ	0.2 / 0.999



Training convergence graphs (TensorBoard)

- Gymnasium-compliant wrapper — any SB3 algorithm can be substituted
- Structured telemetry: SHILATE-METRIC / SHILATE-HEALTH markers parsed live by the Editor
- 5-second observation timeout triggers health alert

```
[ OK ] Started Mosquitto MQTT Broker.
[ OK ] Started Eclipse Manto - Update Manager.
[ OK ] Started containerd container runtime.
[ OK ] Started Eclipse Manto - Container Management.

Eclipse Leda v0.1.0-R3
leda-525400123456 login: root

[leda]

Eclipse Leda v0.1.0-R3 (GNU/Linux 5.15.124-yocto-standard x86_64)

Hostname:..... leda-525400123456
Uptime:..... 0 days, 0:0 hours
System load:..... 0.63 (1min) | 0.15 (5min) | 0.85 (15min)
Disk Space:..... Root: 698.7M | Used: 518.8M | Free: 128.1M
Data: 8.6G | Used: 4.8G | Free: 3.8G
RAM:..... Total: 2.8G | Used: 128.8M | Free: 3.6G
Network:..... not connected or activated

To check SDV services health, run $ sdv-health
Use $ ado-device-info to display device configuration
Use $ sdv-provision to generate self-signed device certificates
```

Eclipse Leda container stack

5-Container Docker Compose Stack

- 1 **Mosquitto** — MQTT broker
- 2 **Kuksa databroker** — VSS signal store
- 3 **mqtt-kuksa-feeder** — MQTT → VSS bridge
- 4 **Velocitas app** — safety monitor
- 5 **leda-controller** — RL inference engine

Signal Chain

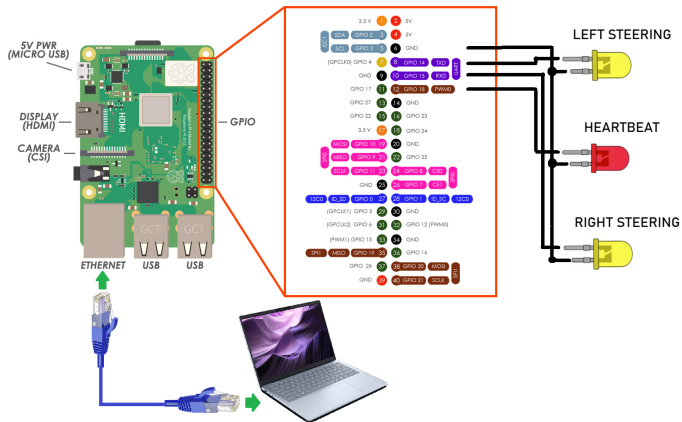
- env0/vehicle/speed → Vehicle.Speed (COVESA VSS)
- Velocitas app: overspeed alert (>120 km/h), RPM alert (>6 500)

Deployment to Raspberry Pi 4

- deploy-pi.sh — cross-compiles ARM64 via docker buildx, transfers and starts stack
- Full deploy cycle: ≈14 minutes

Remote Inference — Training Controller targets the Pi broker; no code changes, policy runs on-device

Hardware Setup — Raspberry Pi 4 Connections



Physical Wiring

- **Ethernet** — direct cable to development laptop running Unity Sim and MQTT broker
- **3 GPIO LEDs** (via UART pins) used as hardware status indicators
- MQTT traffic flows over the LAN link using the same `env0/` topic namespace — zero reconfiguration
- **15–20 ms** round-trip latency measured over direct Ethernet

Key insight

The same containerised inference stack runs identically on the Pi and inside the QEMU Leda image.

Training Convergence (3 seeds, mean)

Milestone	Timestep	σ
First collision-free episode	12 400	$\pm 2\,100$
Half-turn bonus triggered	31 000	$\pm 3\,800$
First full lap	48 500	$\pm 5\,200$
Rolling mean ≥ 60	76 000	$\pm 6\,900$

Inference — 50 episodes (local)

Metric	Value
Lap completion rate	97%
Mean episode reward	63.4
Mean lap duration	38.2 s
Mean speed (completed laps)	47.6 km/h
Collisions per episode	0.06

Ablation Study

- Half-turn bonus removed ($c_h = 0$): first lap delayed by $\approx 18\,000$ steps; rolling mean 12 pts lower at 100K
- Weaker collision penalty ($c_c = -1$): laps completed but agent drove closer to walls

Health Monitoring — All Failure Modes Detected

- Observation timeout (> 5 s) $\rightarrow$ *Critical*
- Python crash $\rightarrow$ non-zero exit code
- MQTT disconnect $\rightarrow$ *Disconnected* within 35 s

Raspberry Pi Remote Inference

Metric	Value
Lap completion rate (LAN)	95%
Additional latency (LAN)	15–20 ms
Deploy time	≈ 14 min

What was achieved

- End-to-end RL → SDV pipeline — same MQTT/VSS protocol from Unity training to Raspberry Pi 4 hardware
- Agent converges within 100 000 timesteps; 97% lap completion in inference
- One-click Training Controller with 5-failure-mode health monitoring
- Validated on physical hardware — 95% remote lap completion over LAN

Future Work

- Domain randomisation for sim-to-real transfer
- Safety monitor → active command override (not just alerts)
- Multi-agent scenarios and cooperative driving
- CAN bus hardware-in-loop testing on real ECUs



Thank you! Questions?